

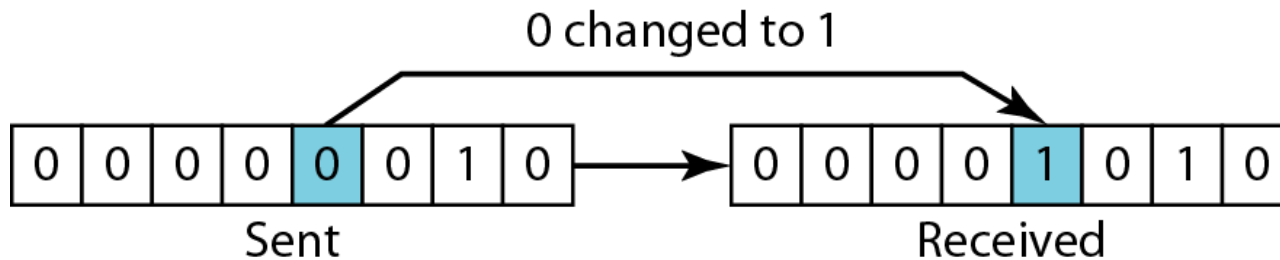
Ramaiah Institute of Technology

Department of AI & DS

Data Communication & Networking

Unit 3

Figure 10.1 *Single-bit error*

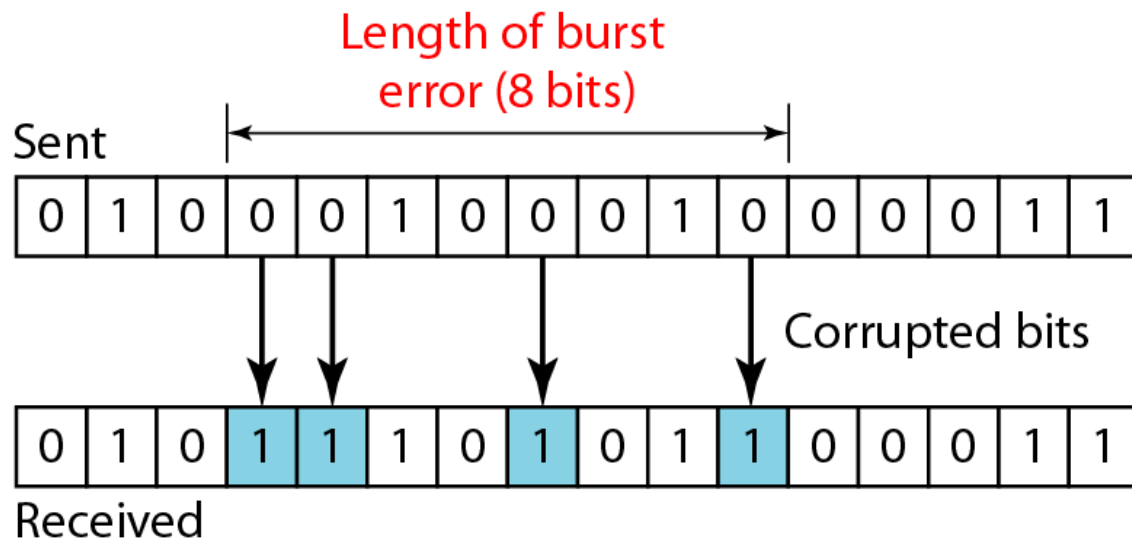


Data can be corrupted during transmission.

Some applications require that errors be detected and corrected.

In single bit error only 1 bit of data is changed

Figure 10.2 *Burst error of length 8*

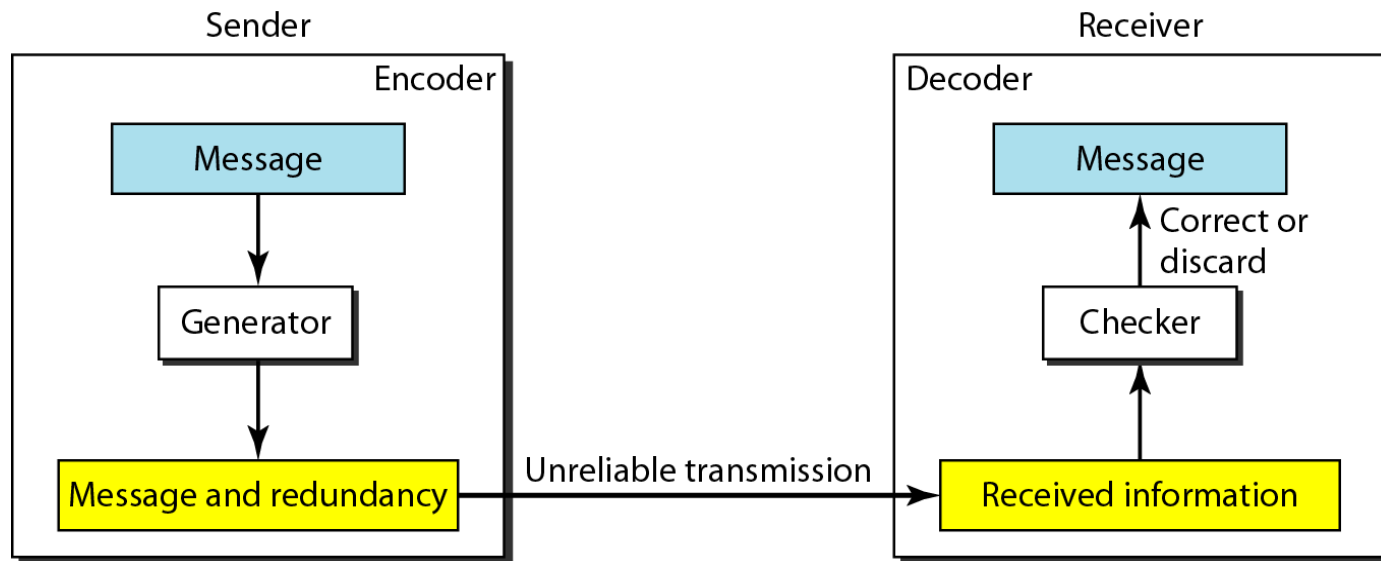


A burst error means that 2 or more bits in the data unit have changed.

To detect or correct errors, we need to send extra (redundant) bits with data.

Figure 10.3 *The structure of encoder and decoder*

To detect or correct errors, we need to send some extra bits with our data. These redundant bits are added by the sender and removed by the receiver.



Detection versus Correction

- **Error Detection:** Check if any error has occurred. The answer is a simple yes or no. We are not even interested in the number of corrupted bits.
- **Error Correction:** Need to know the exact number of bits that are corrupted and, more importantly, their location in the message.
- **The correction of errors is more difficult than the detection.**

Coding

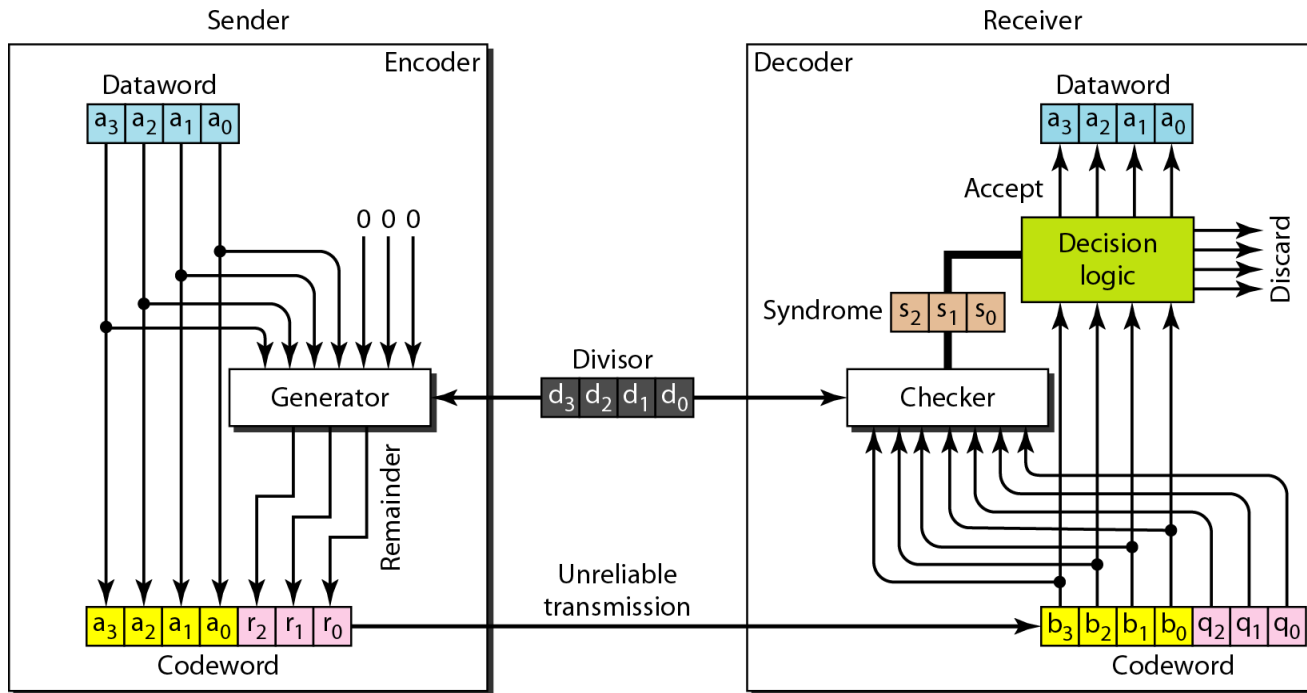
- Redundancy is achieved through various coding schemes.
- The sender adds redundant bits through a process that creates a relationship between the redundant bits and the actual data bits.
- The receiver checks the relationships between the two sets of bits to detect errors.
- The ratio of redundant bits to data bits and the robustness of the process are important factors in any coding scheme.

Cyclic Redundancy Check (CRC)

Table 10.3 A CRC code with $C(7, 4)$

<i>Dataword</i>	<i>Codeword</i>	<i>Dataword</i>	<i>Codeword</i>
0000	0000000	1000	1000101
0001	0001011	1001	1001110
0010	0010110	1010	1010011
0011	0011101	1011	1011000
0100	0100111	1100	1100010
0101	0101100	1101	1101001
0110	0110001	1110	1110100
0111	0111010	1111	1111111

Figure 10.14 *CRC encoder and decoder*



Cyclic Redundancy Check (CRC)

Encoder

- The dataword has k bits (4 here); the codeword has n bits (7 here).
 - The size of the dataword is augmented by adding $n - k$ (3 here) 0s to the right-hand side of the word. The n -bit result is fed into the generator.
 - The generator uses a divisor of size $n - k + 1$ (4 here), predefined and agreed upon.
 - The generator divides the augmented dataword by the divisor (modulo-2 division).
 - The quotient of the division is discarded; the remainder ($r_2r_1r_0$) is appended to the dataword to create the codeword.
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Cyclic Redundancy Check (CRC)

Decoder

- The decoder receives the codeword (possibly corrupted in transition).
 - A copy of all n bits is fed to the checker, which is a replica of the generator.
 - The remainder produced by the checker is a syndrome of $n - k$ (3 here) bits, which is fed to the decision logic analyzer.
 - The analyzer has a simple function. If the syndrome bits are all 0s, the 4 leftmost bits of the codeword are accepted as the dataword (interpreted as no error); otherwise, the 4 bits are discarded (error)
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Figure 10.15 *Division in CRC encoder*

The encoder takes a dataword and augments it with $n - k$ number of 0s. It then divides the augmented dataword by the divisor

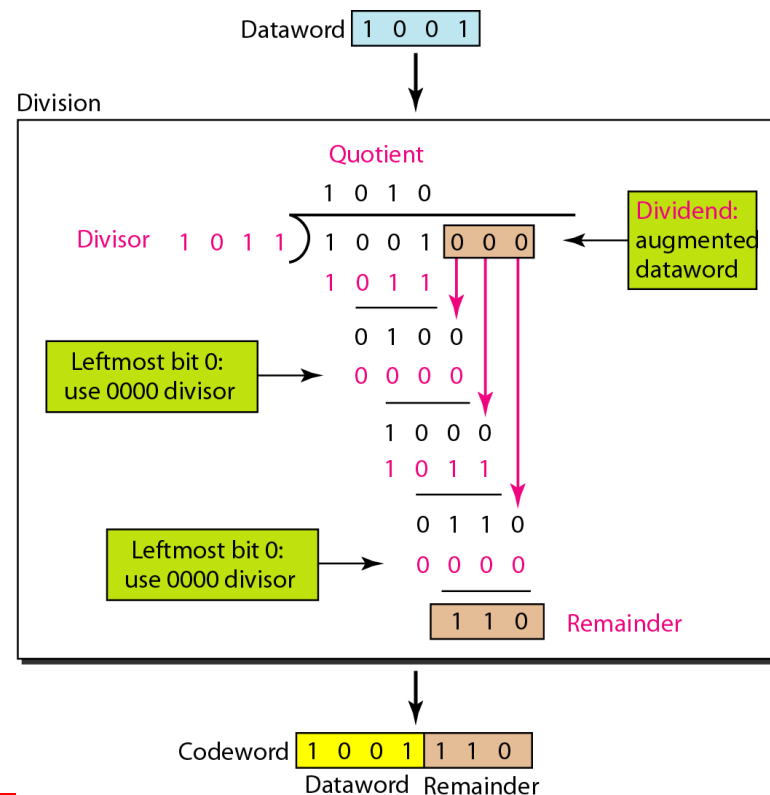
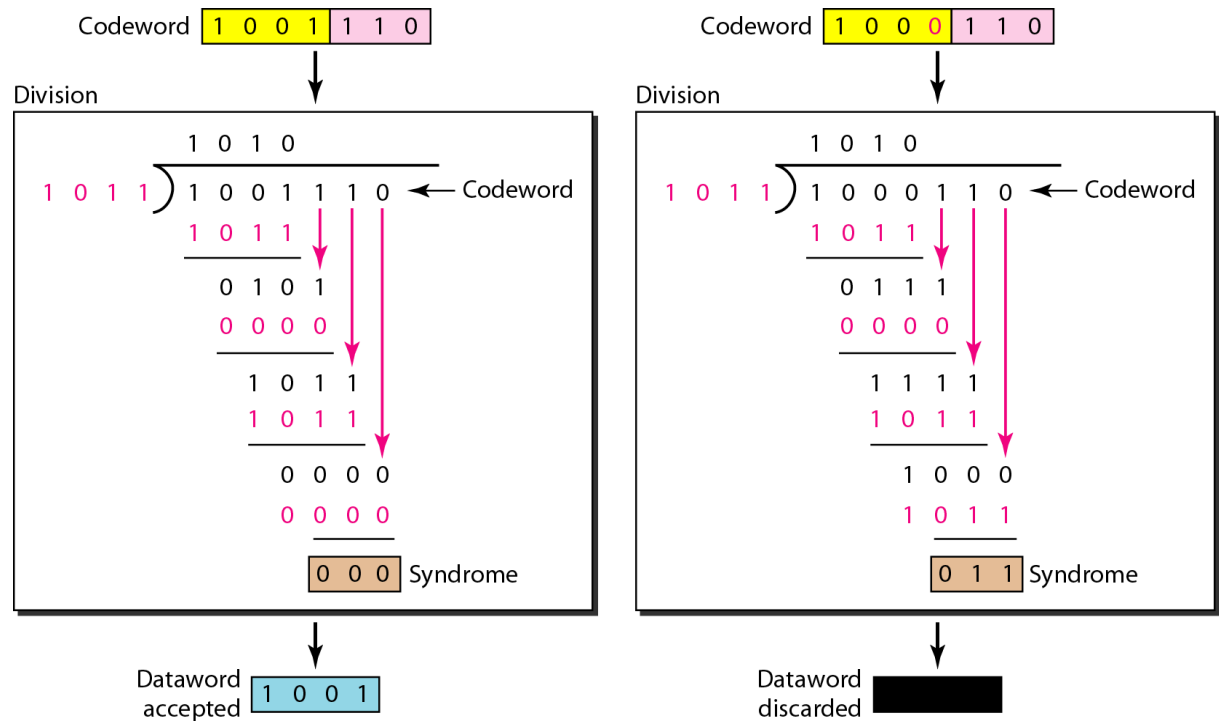


Figure 10.16 *Division in the CRC decoder for two cases*

- The codeword can change during transmission.
- The decoder does the same division process as the encoder. The remainder of the division is the syndrome.
- If the syndrome is all 0s, there is no error with a high probability; the dataword is separated from the received codeword and accepted.
- Otherwise, everything is discarded.



CRC Polynomial

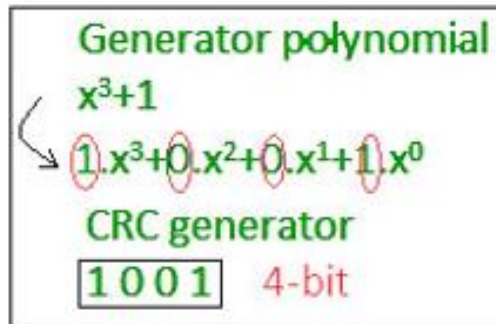
The divisor in a cyclic code is normally called the generator polynomial or simply the generator.

Table 10.4 *Standard polynomials*

<i>Name</i>	<i>Polynomial</i>	<i>Used in</i>
CRC-8	$x^8 + x^2 + x + 1$ 100000111	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$ 11000110101	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$ 10001000000100001	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$ 100000100110000010001110110110111	LANs

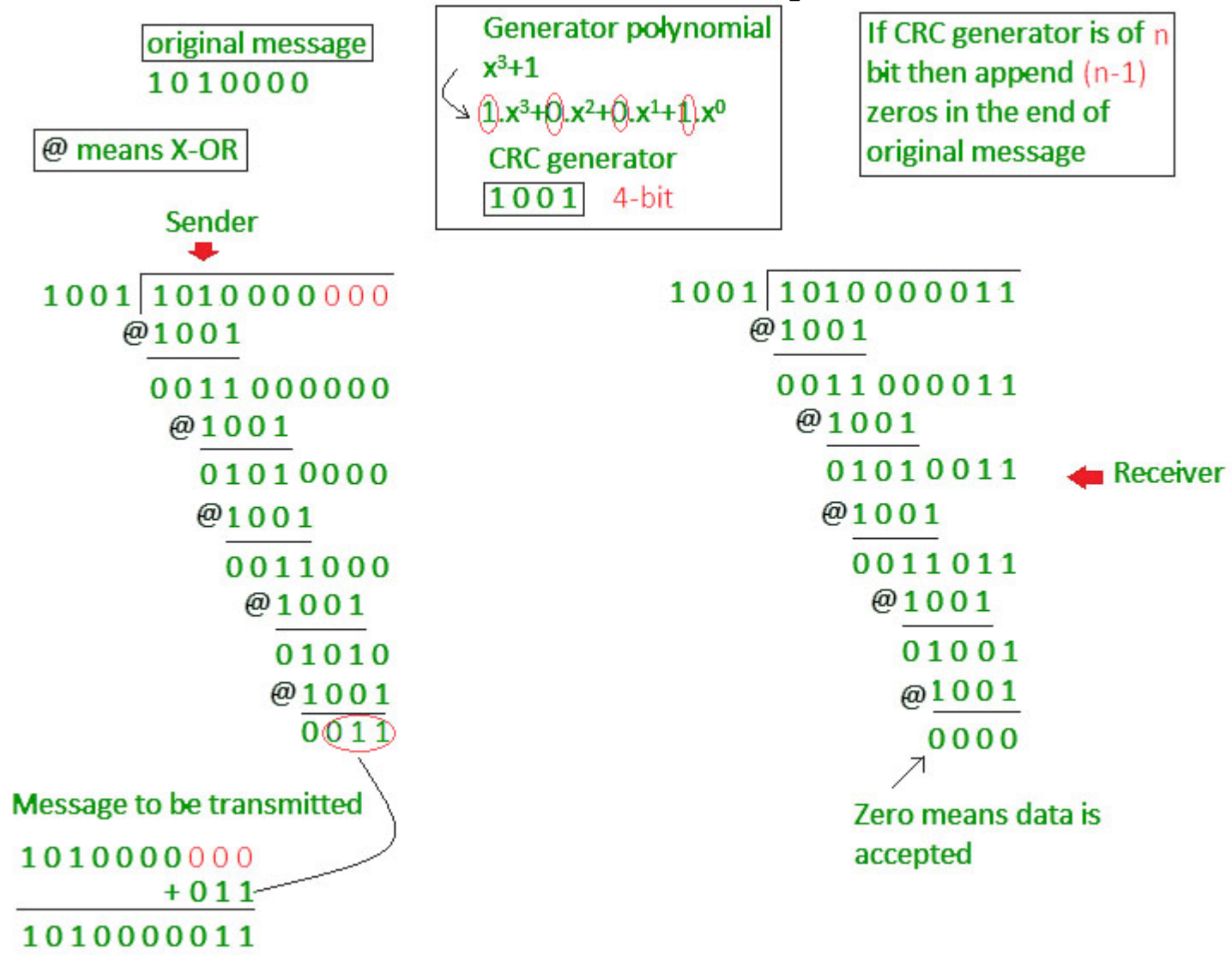
CRC Example

1. Given the dataword 1010000 and the divisor x^3+1 , show the generation of the CRC codeword at the sender site (using binary division) and compute the decoding on receiving side to check errors.



If CRC generator is of n bit then append $(n-1)$ zeros in the end of original message

CRC Example



CRC Example

1. Given the dataword 1001 and the divisor 1011, show the process of encoding and decoding at the sender site and receiver site using polynomial division.
2. Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division) and compute the decoding on receiving side to check errors.
3. A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1 .
 - i. What is the actual bit string transmitted?
 - ii. Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?
4. Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division).